Tal Lavian, Ph.D.

https://TelecommNet.com
tlavian@TelecommNet.com

Encino, CA 91316
(408)-209-9112

Telecommunications, Network Communications, Mobile Wireless, and Internet Technologies Expert

Dr. Lavian is a scientist, educator, and technologist with over 35 years of experience. He has co-authored over 25 scientific publications, journal articles, and peer-reviewed papers. He is an expert in network communications and telecommunications, including Internet protocols, data communications, and computer networks.

Dr. Lavian has spent 20 years researching, studying, and lecturing at UC Berkeley's College of Engineering. His research focuses on telecommunications systems, network communications, network services, network software, network protocols, Internet Protocols TCP/IP, and communications frameworks.

He holds a Ph.D. in Computer Science from UC Berkeley, specializing in network communications; an M.Sc. in Electrical Engineering from Tel Aviv University; and a B.Sc. in Mathematics and Computer Science.

EXPERTISE

Network communications, telecommunications, Internet protocols, and mobile wireless:

- **Network Communications:** Internet protocols; TCP/IP suite, TCP, UDP, IP, Ethernet, 802.3, LAG, network protocols, network software applications, data link, network, transport layers, SNMP, NMS, network management, packet switching, and network architecture.
- **VoIP/Streaming Media:** VoIP, SIP, RTP, video/audio conferencing, streaming media, IP telephony, transport systems, PSTN, circuit switching, SS7, SONET, SDH, and TDM.
- **Mobile wireless:** Wi-Fi, 802.11, Bluetooth, 802.15, Wireless LAN (WLAN), MAC, PHY, ARQ, HARQ. Cellular, SMS, MMS, messaging, chat, mobile devices, and smartphones.
- **Internet/cloud:** Internet Technologies, Web applications, HTTP, HTTPS, cybersecurity, firewalls, SSH, SSL, FTP, e-mail, client-server, cloud computing, and distributed computing.
- **Routing/switching:** LAN, WAN, VPN, encapsulation, routing protocols, RIP, BGP, MPLS, OSPF, multicast, VPLS, Pseudowire, DNS, QoS, queuing, traffic control, network security, deep packet inspection, L4-L7 switching, servers, gateways, network infrastructure, and architectures.

Dr. Lavian has extensive experience in software development for computer networks, architectures, configurations, installations, and network testing. He has academic and hands-on experience in the above fields, including technology products from different companies, implementations, related standards, designs, systems, hardware, and software technologies.

ACCOMPLISHMENTS

- Principal Investigator (PI) for three US Department of Defense (DARPA) projects.
- Directed networking computation project for the US Air Force Research Lab (AFRL).
  PI of a wireless research project for an undisclosed US federal agency.
- An inventor of over 120 patents, over 60 prosecuted *pro-se* before the USPTO.
- Led and developed the first network resource scheduling service for grid computing.

- Managed and engineered the first demonstrated dynamic transatlantic allocation of 10Gbs Lambdas as a grid service.
- Developed and successfully demonstrated the first wire-speed active network on commercial hardware.
- Created and chaired Nortel Networks' EDN Patent Committee.

EXPERT WITNESS

Dr. Lavian has served as an expert witness in cases involving over 140 patents, providing expert reports and testimony in over 70 depositions. He has also testified in Federal courts, before judges and juries, USPTO PTAB IPR, and the ITC. These cases involved leading companies such as Amazon, LinkedIn, Avaya, Netflix, T-Mobile, ZTE, Ericsson, Cisco Systems, Juniper Networks, Arista Networks, Polycom, Motorola, LG, WhatsApp, Instagram, Microsoft, Google, Huawei, Facebook, and Apple.

PROFESSIONAL EXPERIENCE

The University of California, Berkeley, Berkeley, California 2000-2019
U.C. Berkeley SkyDeck, Industry Fellow, Lecturer, Visiting Scientist, Ph.D. Candidate, Nortel's Scientist Liaison
Some positions and projects were concurrent, others sequential.

- U.C. Berkeley SkyDeck startups - advanced technology research, development, business, and market.
- Industry fellow and lecturer at the Sutardja Center for Entrepreneurship and Technology (SCET).
- Conducted research projects in data centers (RAD Labs), telecommunication infrastructure (SAHARA), and wireless systems (ICEBERG).
- Acted as a scientific liaison between Nortel Research Lab and U.C. Berkeley, providing tangible value in advanced technologies.
- Developed long-term technology for the enterprise market, integrating communication and computing technologies.
- Studied network services, telecommunication systems and software, communications infrastructure, and data centers.
- Earned a Ph.D. in Computer Science with a specialization in network communications.

TelecommNet Engineering, Inc. Sunnyvale, California 2006-Present
Principal Scientist

- Consulting in network communications, telecommunications, Internet protocols, and smartphone mobile wireless devices.
- Providing system architecture and technology analysis for computer networks, mobile wireless devices, and Internet web technologies projects.
- Providing expert witness services in network communications patent infringement lawsuits.

CRadar.Ai, U.C. Berkeley, California 2018-2019
CTO / Principal Investigator

- CRadar.Ai improves the Radar wireless RF signal phase noise purity by 100x.

- Accurate Radars are paramount for self-driving car safety. Radars "see" where Cameras and LiDars are "blind" (fog, rain, snow, direct sunlight, and darkness).
- The superior wireless RF signal quality provides a clean signal for high Radar accuracy.
- Improving Radar accuracy and resolution enables genuine redundancy and sensory fusion and puts the Radar into the sensory spearhead.

Aybell (VisuMenu Inc.), U.C. Berkeley, California 2016-Present

CEO/CTO

- Aybell transforms smartphones into visual menu systems, making the phone a frictionless point for user interactions with customer service platform features. Empower consumers to reach suitable agents in call centers, overcoming customer service barriers. Aybell is a brand and marketer of VisuMenu advanced technologies.
- Architecture, design, and implementation of a cloud data center for connecting smartphone users to any company and service by digitizing interactive voice systems and exposing APIs to other applications through cloud service.
- The system was deployed as a cloud networking and cloud computing service on Amazon Web Services (AWS) and Google Cloud Platform (GCP).
- Technologies include Data Science analytics, Machine Learning (ML), Artificial Intelligence (AI), and Statistical Learning (SL). Building an NLP Parser using Python and other NLP libraries and modules.

VisuMenu, Inc., Sunnyvale, California 2010-2016

Co-Founder and Chief Technology Officer (CTO)

- Led the software design and development of a visual IVR system for smartphones and other mobile devices based on the innovative use of wireless and network communications technologies.
- Designed a voice search engine for IVR / PBX using Asterisk, SIP, and VoIP.
- The system was deployed as a cloud networking and cloud computing service on Amazon Web Services (AWS) and Google Cloud Platform (GCP).
- VisuMenu advanced technologies rebranded as Aybell.

Ixia, Santa Clara, California 2008 - 2008

Network Communications Consultant

Researched and developed advanced network communications testing technologies:

- IxNetwork/IxN2X —IP routing, switching devices, and broadband access equipment. Provided traffic generation and emulation for the full range of protocols: OSPF, RIP, EIGRP, BGP, IS-IS, MPLS, unicast, multicast, broadcast, layer 2/3 VPNs, IPSec, carrier Ethernet, broadband access, and data center bridging. Tested and validated compatibility with IEEE, ITU, and IETF RFC standards.
- IxLoad — quickly and accurately modeled high-volume video, data, and voice subscribers and servers to test the real-world performance of multiservice delivery and security platforms.
- IxCatapult emulated a broad range of wireless access and core protocols to test wireless components and systems. When combined with IxLoad, this provides an end-to-end solution for testing wireless service quality.

- IxVeriWave — employed a client-centric model to test Wi-Fi and wireless LAN networks by generating repeatable, large-scale, real-world test scenarios that are virtually impossible to create by any other means.
- Test automation — provided simple, comprehensive lab automation to help test engineering teams create, organize, catalog, and schedule the execution of tests.

Nortel Networks, Santa Clara, California 1996 - 2007

Employed initially by Bay Networks, later acquired by Nortel Networks

Principal Scientist, Principal Architect, Principal Engineer, Senior Software Engineer

Held scientific and research roles at Nortel Labs, Bay Architecture Labs, and the CTO's office.

Principal Investigator for U.S. Department of Defense (DARPA) Projects

- Conceived, proposed, and completed three research projects: active networks, DWDM-RAM, and a networking computation project for the Air Force Research Lab (AFRL).
- Led a wireless research project for an undisclosed U.S. federal agency.

Academic and Industrial Researcher

- Analyzed new technologies to reduce risks associated with R&D investment.
- Led research collaborations with leading universities and professors at UC Berkeley, Northwestern University, the University of Amsterdam, and the University of Technology, Sydney.
- Evaluated competitive products relative to Nortel's products and technology.
- Proactively identified prospective business ideas, leading to new networking products.
- Predicted technological trends through researching the technological horizon and academic sphere.
- Designed software for switches, routers, and network communications devices.
- Developed systems and architectures for switches, routers, and network management.
- Researched and developed the following projects:
  - Data-Center Communications: network and server orchestration 2006-2007
  - DRAC: SOA-facilitated L1/L2/L3 network dynamic controller 2003-2007
  - Omega: classified project for undisclosed U.S. Federal Agency 2006-2006
  - Platform project for the U.S. Air Force Research Laboratory (AFRL) 2005-2005
  - Network resource orchestration for Web services workflows 2004-2005
  - A proxy study between Web/grids services and network services 2004-2004
  - Streaming content replication: real-time A/V media multicast at edge 2003-2004
  - DWDM-RAM: U.S. DARPA-funded program on agile optical transport 2003-2004
  - Packet capturing and forwarding service on IP and Ethernet traffic 2002-2003
  - CO2: content-aware agile networking 2001-2003
  - Active networks: US DARPA-funded research program 1999-2002
  - ORE: programmable network service platform 1998-2002
  - JVM platform: Java on network devices 1998-2001
  - Web-based device management: network device management 1996-1997

Technology Innovator and Patent Leader

- Created and chaired Nortel Networks' EDN Patent Committee.
- Facilitated a continuous stream of innovative ideas and their conversion into intellectual property rights.
- Developed intellectual property assets through invention and analysis of existing technology portfolios.

Aptel Communications, Netanya, Israel 1994-1995

Software Engineer, Team Leader

Start-up company focused on mobile wireless CDMA spread spectrum PCN/PCS.

- Developed a mobile wireless device using an unlicensed band - Direct Sequence Spread Spectrum (DSSS); FCC part 15 - unlicensed transmitters.
- Designed and managed a personal communication network (PCN) and personal communication system (PCS), which were the precursors of short text messages (SMS).
- Designed and developed network communications software products in C/C++.
- Invented and implemented a two-way paging product.

Scitex Ltd., Herzeliya, Israel 1990-1993

Software Engineer, Team Leader

Software and hardware company acquired by Hewlett-Packard (HP)

- Developed system and network communications in C/C++.
- I provided IT services, System Administration, and network administration.
- I worked on Unix systems, including IBM AIX, HP, and SUN Unix.
- Invented Parallel SIMD Architecture.
- Participated in the Technology Innovation group.

Shalev, Ramat-HaSharon, Israel 1987-1990

Start-up company

Software Engineer

- Developed real-time software and algorithms in C/C++ and Pascal.

PROFESSIONAL ASSOCIATIONS

- IEEE senior member
- IEEE CNSV co-chair, Intellectual Property SIG (2013)
- President Next Step Toastmasters (an advanced TM club in the Silicon Valley) (2013-2014)
- Technical co-chair, IEEE Hot Interconnects 2005 at Stanford University
- Member, IEEE Communications Society (COMMSOC)
- Member, IEEE Computer Society
- Member, IEEE Systems, Man, and Cybernetics Society
- Member, IEEE-USA Intellectual Property Committee (2012)
- Member, ACM, ACM Special Interest Group on Data Communication (SIGCOM)
- Member, ACM Special Interest Group on Hypertext, Hypermedia, and Web (SIGWEB)
- Member, IEEE Consultants' Network (CNSV)
- Global Member, Internet Society (ISOC)
- President Java Users Group – Silicon Valley Mountain View, CA,1999-2000
- Toastmasters International

FORMER ADVISORY BOARD POSITIONS

- Quixey –search engine for wireless mobile apps
- Mytopia – mobile wireless social games
- iLeverage – Israeli Innovations

PROFESSIONAL AWARDS

- Top Talent Award – Nortel
- Top Inventors Award – Nortel EDN
- Certified IEEE-WCET - Wireless Communications Engineering Technologies (2012)
- Toastmasters International - Competent Communicator (twice)
- Toastmasters International - Advanced Communicator Bronze
- Best Paper Presentation Award - ICE/IEEE Conference. "R&D Models for Advanced Development & Corporate Research"

PERSONAL

- USA FIT – San Jose Marathon running club (2017-2020)

Patents and Publications

Patents Issued
(Representative List)

US 9,690,877	Systems and methods for electronic communications	Link
US 9,660,655	Ultra-low phase noise frequency synthesizer	Link
US 9,184,989	Grid proxy architecture for network resources	Link
US 9,521,255	Systems and methods for visual presentation and selection of IVR menu	Link
US 9,083,728	Systems and methods to support sharing and exchanging in a network	Link
US 9,021,130	Photonic line sharing for high-speed routers	Link
US 8,762,963	Translation of programming code	Link
US 8,762,962	Methods and apparatus for automatic translation of a computer program language code	Link
US 8,745,573	Platform-independent application development framework	Link
US 8,731,148	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,688,796	Rating system for determining whether to accept or reject an objection raised by a user in a social network	Link
US 8,619,793	Dynamic assignment of traffic classes to a priority queue in a packet-forwarding device	Link
US 8,572,303	A portable Universal communication device	Link
US 8,553,859	Device and method for providing enhanced telephony	Link
US 8,548,131	Systems and methods for communicating with an interactive voice response system	Link
US 8,537,989	Device and method for providing enhanced telephony	Link
US 8,341,257	Grid proxy architecture for network resources	Link
US 8,161,139	Method and apparatus for intelligent management of a network element	Link
US 8,146,090	Time-value curves to provide dynamic QoS for time-sensitive file transfer	Link
US 8,078,708	Grid proxy architecture for network resources	Link
US 7,944,827	Content-aware dynamic network resource allocation	Link

US 7,860,999	Distributed computation in network devices	Link
US 7,734,748	Method and apparatus for intelligent management of a network element	Link
US 7,710,871	Dynamic assignment of traffic classes to a priority queue in a packet-forwarding device	Link
US 7,580,349	Content-aware dynamic network resource allocation	Link
US 7,433,941	Method and apparatus for accessing network information on a network device	Link
US 7,359,993	Method and apparatus for external interfacing resources with a network element	Link
US 7,313,608	Method and apparatus for using documents written in a markup language to access and configure network elements	Link
US 7,260,621	The object-oriented network management interface	Link
US 7,237,012	Method and apparatus for classifying Java remote method invocation transport traffic	Link
US 7,127,526	Method and apparatus for dynamically loading and managing software services on a network device	Link
US 7,047,536	Method and apparatus for classifying remote procedure call transport traffic	Link
US 7,039,724	Programmable command-line interface API for managing the operation of a network device	Link
US 6,976,054	Method and system for accessing low-level resources in a network device	Link
US 6,970,943	Routing architecture includes a compute plane configured for high-speed processing of packets to provide application layer support.	Link
US 6,950,932	Security association mediator for Java-enabled devices	Link
US 6,850,989	Method and apparatus for automatically configuring a network switch	Link
US 6,845,397	Interface method and system for accessing inner layers of a network protocol	Link
US 6,842,781	Download and processing of a network management application on a network device	Link
US 6,772,205	Executing applications on a target network device using a proxy network device	Link
US 6,564,325	Method of and apparatus for providing multi-level security access to a system	Link
US 6,175,868	Method and apparatus for automatically configuring a network switch	Link
US 6,170,015	Network apparatus with Java co-processor	Link
US 8,687,777	Systems and methods for visual presentation and selection of IVR menu	Link

US 8,681,951	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,625,756	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,594,280	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,548,135	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,406,388	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,345,835	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,223,931	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,160,215	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,155,280	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,054,952	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,000,454	Systems and methods for visual presentation and selection of IVR menu	Link
EP 1,905,211	A technique for authenticating network users	Link
EP 1,142,213	Dynamic assignment of traffic classes to a priority queue in a packet forwarding device	Link
EP 1,671,460	Method and apparatus for scheduling resources on a switched underlay network	Link
US 9,001,819	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,949,846	Time-value curves to provide dynamic QoS for time-sensitive file transfers	Link
US 8,929,517	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,903,073	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,898,274	Grid proxy architecture for network resources	Link
US 8,880,120	Device and method for providing enhanced telephony	Link
US 8,879,703	System method and device for providing tailored services when a call is on-hold	Link
US 8,879,698	Device and method for providing enhanced telephony	Link
US 8,867,708	Systems and methods for visual presentation and selection of IVR menu	Link
US 8,787,536	Systems and methods for communicating with an interactive voice response system	Link
US 8,782,230	Method and apparatus for using a command design pattern to access and configure network elements	Link

CA 2,358,525	Dynamic assignment of traffic classes to a priority queue in a packet-forwarding device	Link
CA 2,989,752	Ultra-low Phase Noise Frequency Synthesizer	Link
US 10,598,764	Radar target detection and imaging system for autonomous vehicles with ultra-low phase noise frequency synthesizer	Link
US 10,404,261	Radar target detection system for autonomous vehicles with an ultra-low phase-noise frequency synthesizer	Link
US 10,348,313	Radar target detection system for autonomous vehicles with an ultra-low phase-noise frequency synthesizer	Link
US 10,205,457	RADAR target detection system for autonomous vehicles with an ultra-low phase-noise frequency synthesizer	Link
US 10,764,264	Technique for authenticating network users	Link
EP 3,311,493	An ultra-low phase-noise frequency synthesizer	Link
US 9,831,881	Radar target detection system for autonomous vehicles with ultra-low phase noise frequency synthesizer	Link
US 9,762,251	Ultra-low phase noise frequency synthesizer	Link
US 9,705,511	Ultra-low phase noise frequency synthesizer	Link

Patent Applications Published and Pending
(Representative List)

US 20150058490	Grid Proxy Architecture for Network Resources	Link
US 20150010136	Systems and Methods for Visual Presentation and Selection of IVR Menu	Link
US 20140379784	Method and Apparatus for Using a Command Design Pattern to Access and Configure Network Elements	Link
US 20140105025	Dynamic Assignment of Traffic Classes to a Priority Queue in a Packet Forwarding Device	Link
US 20140105012	Dynamic Assignment of Traffic Classes to a Priority Queue in a Packet Forwarding Device	Link
US 20140012991	Grid Proxy Architecture for Network Resources	Link
US 20130080898	Systems and Methods for Electronic Communications	Link
US 20130022191	Systems and Methods for Visual Presentation and Selection of IVR Menu	Link
US 20130022183	Systems and Methods for Visual Presentation and Selection of IVR Menu	Link
US 20130022181	Systems and Methods for Visual Presentation and Selection of IVR Menu	Link
US 20120180059	Time-Value Curves to Provide Dynamic QoS for Time Sensitive File Transfers	Link
US 20120063574	Systems and Methods for Visual Presentation and Selection of IVR Menu	Link
US 20110225330	Portable Universal Communication Device	Link
US 20100220616	Optimizing Network Connections	Link
US 20100217854	Method and Apparatus for Intelligent Management of a Network Element	Link
US 20100146492	Translation of Programming Code	Link
US 20100146112	Efficient Communication Techniques	Link
US 20100146111	Efficient Communication in a Network	Link
US 20090313613	Methods and Apparatus for Automatic Translation of a Computer Program Language Code	Link
US 20090313004	Platform-Independent Application Development Framework	Link
US 20090279562	Content-aware dynamic network resource allocation	Link
US 20080040630	Time-Value Curves to Provide Dynamic QoS for Time Sensitive File	Link

	Transfers	
US 20070169171	A technique for authenticating network users	Link
US 20060123481	Method and apparatus for network immunization	Link
US 20060075042	Extensible Resource Messaging Between User Applications and Network Elements in a Communication Network	Link
US 20050083960	Method and Apparatus for Transporting Parcels of Data Using Network Elements with Network Element Storage	Link
US 20050076339	Method and Apparatus for Automated Negotiation for Resources on a Switched Underlay Network	Link
US 20050076336	Method and Apparatus for Scheduling Resources on a Switched Underlay Network	Link
US 20050076173	Method And Apparatus for Preconditioning Data to Be Transferred on a Switched Underlay Network	Link
US 20050076099	Method and Apparatus for Live Streaming Media Replication in a Communication Network	Link
US 20050074529	Method and apparatus for transporting visualization information on a switched underlay network	Link
US 20040076161	Dynamic Assignment of Traffic Classes to a Priority Queue in a Packet Forwarding Device	Link
US 20020021701	Dynamic Assignment of Traffic Classes to a Priority Queue in a Packet Forwarding Device	Link
WO 2006/063052	Method and apparatus for network immunization	Link
WO 2007/008976	A technique for authenticating network users	Link
WO2000/054460	Method and apparatus for accessing network information on a network device	Link
WO/2016/203460	Ultra-low phase noise frequency synthesizer	Link
WO/2005/033899	Method and apparatus for scheduling resources on a switched underlay network	Link
WO/2000/041368	Dynamic assignment of traffic classes to a priority queue in a packet forwarding device	Link
US 20140156556	A Time-variant rating system and method thereof	Link
US 20140156758	A Reliable rating system and method thereof	Link

US 20170085708	Systems and methods for visual presentation and selection of IVR menu	Link
US 20160373117	Ultra-low phase noise frequency synthesizer	Link
US 20170322687	Systems and methods for electronic communications	Link
US 20170302282	Radar target detection system for autonomous vehicles with ultra-low phase noise frequency synthesizer	Link
US 20180019755	Radar target detection system for autonomous vehicles with ultra-low phase noise frequency synthesizer	Link
US 20170289332	Systems and methods for visual presentation and selection of IVR menu	Link
US 20170269797	Systems and methods for electronic communication	Link
US 20170099058	Ultra-low phase noise frequency synthesizer	Link
US 20170099057	Ultra-low phase noise frequency synthesizer	Link
US 20190128998	Radar target detection and imaging system for autonomous vehicles with ultra-low phase noise frequency synthesizer	Link
US 20190082043	Systems and methods for visual presentation and selection of IVR menu	Link
US 20180146090	Systems and methods for visual presentation and selection of IVR menu	Link
US 20180130102	Reliable rating system and method thereof	Link

Publications
(Representative List)

- "R&D Models for Advanced Development & Corporate Research" Understanding Six Models of Advanced R&D - Ikhlaq Sidhu, Tal Lavian, Victoria Howell - University of California, Berkeley. ASEE Annual Conference and Exposition- 2015. Received "Best Paper Presentation Award" ICE/IEEE Conference June 2015.
- "Communications Architecture in Support of Grid Computing," Tal Lavian, Scholar's Press 2013ISBN 978-3-639-51098-0.
- "Applications Drive Secure Light-path Creation across Heterogeneous Domains, Feature Topic Optical Control Planes for Grid Networks: Opportunities, Challenges, and the Vision." Gommans L.; Van Oudenaarde B.; Dijkstra F.; De Laat C.; Lavian T.; Monga I.; Taal A.; Travostino F.; Wan A.; IEEE Communications Magazine, vol. 44, no. 3, March 2006, pp. 100-106.
- *Lambda Data Grid: Communications Architecture in Support of Grid Computing*. Tal I. Lavian, Randy H. Katz; Doctoral Thesis, University of California at Berkeley. January 2006.
- "Information Switching Networks." Hoang D.B.; T. Lavian; The 4th Workshop on the Internet, *Telecommunications and Signal Processing, WITSP2005*, December 19-21, 2005, Sunshine Coast, Australia.
- "Impact of Grid Computing on Network Operators and HW Vendors." Allcock B.; Arnaud B.; Lavian T.; Papadopoulos P.B.; Hasan M.Z.; Kaplow W.; *IEEE Hot Interconnects at Stanford University 2005*, pp.89-90.
- *DWDM-RAM: A Data Intensive Grid Service Architecture Enabled by Dynamic Optical Networks*. Lavian T.; Mambretti J.; Cutrell D.; Cohen H.J; Merrill S.; Durairaj R.; Daspit P.; Monga I.; Naiksatam S.; Figueira S.; Gutierrez D.; Hoang D.B., Travostino F.; *CCGRID 2004*, pp. 762-764.
- *DWDM-RAM: An Architecture for Data-Intensive Service Enabled by Next Generation Dynamic Optical Networks*. Hoang D.B.; Cohen H.; Cutrell D.; Figueira S.; Lavian T.; Mambretti J.; Monga I.; Naiksatam S.; Travostino F.; Proceedings IEEE Globecom 2004, Workshop on High-Performance Global Grid Networks, Houston, 29 Nov. to 3 Dec. 2004, pp.400-409.
- *Implementation of a Quality of Service Feedback Control Loop on Programmable Routers*. Nguyen C.; Hoang D.B.; Zhao, I.L.; Lavian, T.; Proceedings, 12th IEEE International Conference on Networks 2004. (ICON 2004) Singapore, Volume 2, 16-19 Nov. 2004, pp.578-582.
- *A Platform for Large-Scale Grid Data Service on Dynamic High-Performance Networks*. Lavian T.; Hoang D.B.; Mambretti J.; Figueira S.; Naiksatam S.; Kaushil N.; Monga I.; Durairaj R.; Cutrell D.; Merrill S.; Cohen H.; Daspit P.; Travostino F; GridNets 2004, San Jose, CA., October 2004.
- *DWDM-RAM: Enabling Grid Services with Dynamic Optical Networks.* Figueira S.; Naiksatam S.; Cohen H.; Cutrell D.; Daspit, P.; Gutierrez D.; Hoang D. B.; Lavian T.; Mambretti J.; Merrill S.; Travostino F; Proceedings, 4th IEEE/ACM International Symposium on Cluster Computing and the Grid, Chicago, USA, April 2004, pp. 707-714.

- *DWDM-RAM: Enabling Grid Services with Dynamic Optical Networks.* Figueira S.; Naiksatam S.; Cohen H.; Cutrell D.; Gutierrez D.; Hoang D.B.; Lavian T.; Mambretti J.; Merrill S.; Travostino F.; 4th IEEE/ACM International Symposium on Cluster Computing and the Grid, Chicago, USA, April 2004.

- *An Extensible, Programmable, Commercial-Grade Platform for Internet Service Architecture.* Lavian T.; Hoang D.B.; Travostino F.; Wang P.Y.; Subramanian S.; Monga I.; IEEE Transactions on Systems, Man, and Cybernetics on Technologies Promoting Computational Intelligence, Openness and Programmability in Networks and Internet Services Volume 34, Issue 1, Feb. 2004, pp.58-68.
- *DWDM-RAM: An Architecture for Data-Intensive Service Enabled by Next Generation Dynamic Optical Networks*. Lavian T.; Cutrell D.; Mambretti J.; Weinberger J.; Gutierrez D.; Naiksatam S.; Figueira S.; Hoang D. B.; Supercomputing Conference, SC2003 Igniting Innovation, Phoenix, November 2003.
- *Edge Device Multi-Unicasting for Video Streaming.* Lavian T.; Wang P.; Durairaj R.; Hoang D.; Travostino F.; Telecommunications, 2003. ICT 2003. 10th International Conference on Telecommunications, Tahiti, Volume 2, 23 Feb.-1 March 2003 pp. 1441-1447.
- The SAHARA Model for Service Composition Across Multiple Providers. Raman B.; Agarwal S.; Chen Y.; Caesar M.; Cui W.; Lai K.; Lavian T.; Machiraju S.; Mao Z. M.; Porter G.; Roscoe T.; Subramanian L.; Suzuki T.; Zhuang S.; Joseph A. D.; Katz Y.H.; Stoica I.; Proceedings of the First International Conference on Pervasive Computing. ACM Pervasive 2002, pp. 1-14.
- *Enabling Active Flow Manipulation in Silicon-Based Network Forwarding Engines.* Lavian T.; Wang P.; Travostino F.; Subramanian S.; Duraraj R.; Hoang D.B.; Sethaput V.; Culler D.; Proceeding of the Active Networks Conference and Exposition, 2002. (DANCE) 29-30 May 2002, pp. 65-76.
- *Practical Active Network Services within Content-Aware Gateways.* Subramanian S.; Wang P.; Durairaj R.; Rasimas J.; Travostino F.; Lavian T.; Hoang D.B.; Proceeding of the DARPA Active Networks Conference and Exposition, 2002. (DANCE) 29-30 May 2002, pp. 344-354.
- *Active Networking on a Programmable Network Platform.* Wang P.Y.; Lavian T.; Duncan R.; Jaeger R.; Fourth IEEE Conference on Open Architectures and Network Programming (OPEN ARCH), Anchorage, April 2002.
- *Intelligent Network Services through Active Flow Manipulation.* Lavian T.; Wang P.; Travostino F.; Subramanian S.; Hoang D.B.; Sethaput V.; IEEE Intelligent Networks 2001 Workshop (IN2001), Boston, May 2001.
- *Intelligent Network Services through Active Flow Manipulation.* Lavian T.; Wang P.; Travostino F.; Subramanian S.; Hoang D.B.; Sethaput V.; Intelligent Network Workshop, 2001 IEEE 6-9 May 2001, pp.73 -82.
- *Enabling Active Flow Manipulation in Silicon-based Network Forwarding Engine.* Lavian, T.; Wang, P.; Travostino, F.; Subramanian S.; Hoang D.B.; Sethaput V.; Culler D.; Journal of Communications and Networks, March 2001, pp.78-87.
- *Active Networking on a Programmable Networking Platform.* Lavian T.; Wang P.Y.; IEEE Open Architectures and Network Programming, 2001, pp. 95-103.

- *Enabling Active Networks Services on a Gigabit Routing Switch.* Wang P.; Jaeger R.; Duncan R.; Lavian T.; Travostino F.; 2nd Workshop on Active Middleware Services, 2000.
- *Dynamic Classification in Silicon-Based Forwarding Engine Environments.* Jaeger R.; Duncan R.; Travostino F.; Lavian T.; Hollingsworth J.; Selected Papers. 10th IEEE Workshop on Metropolitan Area and Local Networks, 1999. 21-24 Nov. 1999, pp.103-109.
- *Open Programmable Architecture for Java-Enabled Network Devices.* Lavian, T.; Jaeger, R. F.; Hollingsworth, J. K.; IEEE Hot Interconnects Stanford University, August 1999, pp. 265-277.
- Open Java SNMP MIB API. Rob Duncan, Tal Lavian, Roy Lee, Jason Zhou, Bay Architecture Lab Technical Report TR98-038, December 1998.
- Java-Based Open Service Interface Architecture. Lavian T.; Lau S.; BAL TR98-010 Bay Architecture Lab Technical Report, March 1998.
- Parallel SIMD Architecture for Color Image Processing. Lavian T. Tel – Aviv University, Tel – Aviv, Israel, November 1995.
- *Grid Network Services, Draft-ggf-ghpn-netservices-1.0.* George Clapp, Tiziana Ferrari, Doan B. Hoang, Gigi Karmous-Edwards, Tal Lavian, Mark J. Leese, Paul Mealor, InderMonga, Volker Sander, Franco Travostino, Global Grid Forum(GGF).
- *Project DRAC: Creating an applications-aware network.* Travostino F.; Keates R.; Lavian T.; Monga I.; Schofield B.; Nortel Technical Journal, February 2005, pp. 23-26.
- *Optical Network Infrastructure for Grid, Draft-ggf-ghpn-opticalnets-1.* Dimitra Simeonidou, Reza Nejabati, Bill St. Arnaud, Micah Beck, Peter Clarke, Doan B. Hoang, David Hutchison, Gigi Karmous-Edwards, Tal Lavian, Jason Leigh, Joe Mambretti, Volker Sander, John Strand, Franco Travostino, Global Grid Forum(GGF) GHPN Standard GFD-I.036 August 2004.
- *Popeye - Using Fine-grained Network Access Control to Support Mobile Users and Protect Intranet Hosts.* Mike Chen, Barbara Hohlt, Tal Lavian, December 2000.
- Open Networking - Better Networking through Programmability, Open Networking - Better Networking through Programmability
- Dangerous Liaisons – Software Combinations as Derivative Works?
  Determann L.; Berkeley Technology Law Journal. Volume 21, Issue 4, Fall 2006. (Lavian T. contributor to the technical section).
- Open Networking – Networking Programmability
  Nortel Seminar, Tal Lavian, August 2000.

Presentations and Talks
(Not an exhaustive list)

- Lambda Data Grid
- A Platform for Large-Scale Grid Data Service on Dynamic High-Performance Networks
- Lambda Data Grid: An Agile Optical Platform for Grid Computing and Data-intensive Applications.
- Workflow Integrated Network Resource Orchestration
- DWDM-RAM: DARPA-Sponsored Research for Data-Intensive Service-on-Demand Advanced Optical Networks
- Impact of Grid Computing on Network Operators and HW Vendors
- Web Services and OGSA
- WINER Workflow Integrated Network Resource Orchestration.
- A Grid Proxy Architecture for Network Resources
- Technology & Society
- Abundant Bandwidth and how it affects us?
- Active Content Networking (ACN)
- DWDM-RAM: Enabling Grid Services with Dynamic Optical Networks
- Application-engaged Dynamic Orchestration of Optical Network Resources
- DWDM-RAM: DARPA-Sponsored Research for Data-Intensive Service-on-Demand Advanced Optical Networks
- An Architecture for Data-Intensive Service Enabled by Next Generation Optical Networks
- A Platform for Data-Intensive Services Enabled by Next Generation Dynamic Optical Networks
- A Platform for Data-Intensive Services Enabled by Next Generation Dynamic Optical Networks
- Optical Networks
- Grid Optical Network Service Architecture for Data-Intensive Applications
- Optical Networking & DWDM
- OptiCal Inc.
- OptiCal & LUMOS Networks
- Optical Networking Services
- Optical Networks
- Business Models for Dynamically Provisioned Optical Networks
- Business Model Concepts for Dynamically Provisioned Optical Networks
- Optical Networks Infrastructure
- Research Challenges in Agile Optical Networks
- •
- Impact on Society
- Technology & Society
- TeraGrid Communication and Computation
- Unified Device Management via Java-enabled Network Devices
- Active Network Node in Silicon-Based L3 Gigabit Routing Switch
- Enabling Active Flow Manipulation (AFM) in Silicon-based Network Forwarding Engines
-

- Active Nets Technology Transfer through High-Performance Network Devices
- Enabling Active Network Services on A Gigabit Routing Switch
- Programmable Network Node: Applications
- Open Innovation via Java-enabled Network Devices
- Practical Considerations for Deploying a Java Active Networking Platform
- Open Programmable Architecture for Java-enabled Network Devices
- Enabling Active Flow Manipulation In Silicon-based Network Forwarding Engines
- Enabling Active Flow Manipulation In Silicon-based Network Forwarding Engines
- Enabling Active Flow Manipulation In Silicon-based Network Forwarding Engines
- DWDM-RAM: DARPA-Sponsored Research for Data-Intensive Service-on-Demand Advanced Optical Networks
- DWDM-RAM: DARPA-Sponsored Research for Data-Intensive Service-on-Demand Advanced Optical Networks
- Open Programmable Architecture for Java-enabled Network Devices
- Open Java-based Intelligent Agent Architecture for Adaptive Networking Devices
- Edge Device Multi-unicasting for Video Streaming
- Intelligent Network Services through Active Flow Manipulation
- Java SNMP Oplet
- Unified Device Management via Java-enabled Network Devices
- Dynamic Classification in a Silicon-Based Forwarding Engine
- Integrating Active Networking and Commercial-Grade Routing Platforms
- Enabling Active Flow Manipulation In Silicon-based Network Forwarding Engines
- Open Distributed Networking Intelligence: A New Java Paradigm
- Open Networking: Better Networking Through Programmability
- Open Programmability
- Active On A Programmable Networking Platform
- Open Networking through Programmability
- Open Programmable Architecture for Java-enabled Network Devices
- Popeye – Fine-grained Network Access Control for Mobile Users
- Integrating Active Networking and Commercial-Grade Routing Platforms
- Active Networking
- Programmable Network Devices
- Open Programmable Architecture for Java-enabled Network Devices
- To be smart or not to be?

Litigation Support Experience - Tal Lavian, Ph.D.

https://TelecommNet.com tlavian@telecommnet.com (408) 209-9112

Case Name: Salutica Allied Solutions v. **Apple** Malaysia
Case: APL2022048377 (Malaysia)
Client: Wong & Partners, Apple's US attorneys Goldman Ismail Baum LLP
Patent No: MY 172803,
Matter: Prepared invalidity expert report. Prepared non-infringement expert report. Bluetooth technology in Apple iPhone, iPad, iWatch, AirPods, and MacBook. Bluetooth IEEE 802.15 standard. January 2023. **Court testimony at trial** 2024

Case Name: Fleet Connect Solutions LLC v. **Teletrac Navman** US Ltd.
Client: Fox Rothschild, LLP
Patent No: US-7,742,388, US-7,593,751, US-6,961,586, US-8,494,581, US- 7,206,837, US-7,463,896, US-7,656,845, and US-7,783,304
Matter: No. 8:23-cv-01759-JWH(DFMx) (C.D. Cal.), *Fleet Connect Solutions LLC v. Teletrac Navman US Ltd*., No. 2:24-cv-00939-JWH(DFMx) (C.D. Cal.)
Re: Infringement, invalidity of network communication hand-held devices
May 2024.

Case Name: **Juniper Networks** v. Portsmouth Networks Corporation
Client: Wolf Greenfield LLP
Patent No: US 8,014,394
Matter: PTAB Declarations re: High-Speed processing for multicast content requests, switch architecture, routing processing design, and multicast traffic over a communication network.
April 2024.

Name: **Unified Patent LLC** v. e-Beacon LLC
Client: Unified Patent LLC.
Patent No: US 8,515,386
Matter: PTAB Declarations re: VoIP Location determination, E-VoIP, PSTN.
2024.

Case Name: Subrigo Communications Corporation v. **Turner Construction**
Case: 21STCV26285
Client: Wood Smith Henning and Berman LLP
Matter: Telecommunications Service Provider infrastructure, reliability, resiliency, and services as a result of a fiber optic cut. 2024

Case Name: **Fortinet,** Inc. v. Forescout Technologies Inc.,
Case No. **20-cv-03343-EMC (JSC)** (Northern District of California)
Client: Skadden, Arp, Slate. Meagher & Flom LLP.

Patent No: Multiple NAC patents.
Matter: Prepared several invalidity, infringement, and rebuttal expert reports and depositions on network security, network access control, security architectures, and access devices.
December 2023.

Case Name: Corrigent Corp. v. **Arista Networks**, Inc.,
Case No. 1:22-cv-00497-UNA (District of Delaware)
USPTO PTAB IPRs
Client: Baker Botts LLP.
Patent No: US 7,593,400, US 7,545,740, 6,957,369, US 9,118,602, and US 7,330,431
Matter: Prepared several IPR invalidity declarations. L2 MAC, VPLS, MPLS, Ring Network topology IEEE 803.17 standard, Spanning Trees IEEE 802.1d standard. 802.3ad Link Aggregation Group (LAG); Switch Architecture; December 2022

Case Name: **Yoo** v. VoIP Guardian, LLC
Central District of California 2:21-ap-01044-BR
Client: BG Law LLP
Matter: Voice over IP CDR. Research and investigation of VoIP CDRs, carrier telephony systems, and architectures. Tier 1, Tier 2, and Tier 3 telephony carriers' interactions and VoIP protocols. May 2023

Case Name: **Sectra Communications AB** v. Absolute Software and NetMotion Software. Case No. 2:22-cv-00353-RSM (W.D. Wash.)
Client: McDermott Will & Emery LLP
Patent Nos: US 7,797,437
Matter: Wireless handover between heterogeneous communications networks.
March 2023

Case Name: **Miami International Securities Exchange**, LLC v. NASDAQ Inc.
USPTO PTAB Reexam preparations
Client: Reed Smith, LLP
Patent Nos: US 7,921,051, US 6,618,707,
Matter: Electronic-based security trading, a configurable computation look-up table. December 2022

Case Name: **Unified Patents** v. 2Bcom LLC.
USPTO PTAB Reexam preparations
Client: Greenberg Traurig, LLP
Patent No: US 7,020,467
Matter: Research and analysis and invalidity claim chart. Bluetooth and wireless channels. Not filed. March 2023

Case Name: **Canada Revenue Services** v. Iris Technologies Inc.
Tax Court Canada case #2021-226(GST)G and 2021-2215(GST)G
Client: Canada Department of Justice

Voice over Internet Protocol (VoIP) technologies, CDR
Matter: Research and analysis of VoIP CDRs. January 2023

Case Name: **Jerry Daniels** v. Gabriela Hernandez Carmona.
civil action - 330h District Court, Dallas County, Texas, DF-22-18099
Court testimony at trial
Client: Law Office of Irene G Mugambi – Dallas, Texas
Telephony CDR, Cellphone tower signals, cellphone locations
Matter: Research and analysis of telephony CDRs - May 2023

Case Name: MOV-Ologi LLC v. **Big Commerce Inc**
Western Texas Case 6:22-cv-00084-ADA
Client: Fish Richardson P.C.
Patent Nos: US 9,286,282 and US 10,769,358
Matter: Claim Constructions. E-commerce, obtaining data from an abandoned webpage form. Declaration August 2022

Case Name: **NetScout Systems** Inc. v. Sons of Innovation Solutions LLC
Case 3:21-cv-1788 (S.D. Cal.) and IPR
Client: EriseIP LLP
Patent Nos: US 8,065,399, US 9,436,542
Matter: Automated network diagnostics. Declaration October 2022

Case Name: **Ericsson** Inc. and **Oracle** Corp v. Telecom Network Solutions LLC
USPTO PTAB IPR
Client: Venable LLP
Patent No: RE47,813
Matter: Feedback Loop for Dynamic Network Resource Allocation as disclosed in standard bodies such as IEEE, IETF RFCs, and 3GPP. September 2022

Case Name: **George Ku** v. Harald Herchen
CA Case No. 21-CV-376210
Client: Farlin, Hecht & Davis LLP
Court testimony at a Jury trial, July 2025
Matter: Google Gmail account and emails from iPhone and Apple MacBook laptop. iPhone, Samsung, and BlackBerry cellphone locations. Email headers analysis, SMTP, and IP locations. 2022.

Case Name: Proven Networks LLC. v. **SAP America, Inc ("SAP")**
C.A. No. 2:21-cv-00100-JRG
Client: Venable LLP
Patent No: US 8,266,530
Matter: Network Management System (NMS), multiple displays of dynamic alarm windows. Claim Construction, Invalidity, Non-Infringement, and PTAB IPR, 2021.

Case Name: **Firtiva Corp** v. Funimation Global Group, LLC,
Texas Eastern District Court – Marshal No. 2:21-cv-111-JRG-RSP
Client: Saul Ewing Arnstein & Lehr LLP
Patent No: US 10,116,999
Matter: A Claim Construction Declaration re: Broadcast communication - embedded information and content in electronic transmission, September 2021.

Case Name: **Juniper Networks** v. Smart Path Connections LLC PTAB
Client: Fisch Sigler LLP
Patent No: US 7,463,580, US 7,386,010, US 7,697,525, 7,961,755
Matter: PTAB Declarations re: MPLS Tunneling, IETF RFCs standards for MPLS, Multiprotocol media conversations, Pseudowire, MAC, SONET, 2021.

Case Name: **Microsoft** v. Uniloc - PTO Ex-Parte Re-examination' '622 patent
Client: Klarquist Sparkman, LLP
Patent No: US. 8,724,622
Matter: An Expert Declaration re: Network Communications, instant VoIP messaging using mobile wireless devices. (Same patent for other parties, Apple, Facebook, and WhatsApp), April 2021.

Case Name: **ZTE Corp.** PTAB IPR
Client: Finnegan, Henderson, Farabow, Garrett & Dunner, LLP
Patent No: US 8,179,960
Matter: An Expert Declaration re: Video communications, streaming, and coding, February 2021.

Case Name: Optis Wireless Technology, LLC, v. **Tesla Inc**.
Client: Munger, Tolles & Olson LLP
Patent No: US 8,149,727
Matter: An Expert consulting re: cellular radio communications and wireless for Tesla electric cars, February 2021

Case Name: **Unified Patents** v. BCS Software LLC. USPTO Reexamination
Client: Unified Patents, LLC
Patent No: US 7,302,612
Matter: An Expert Declaration re: Network Management, Network Health, and operation. Operational Support System (OSS), SNMP, and Device Management, March 2021.

Case Name: WSOU Investment v. **Huawei Technologies**
Client: McGuireWoods,
Patent No: US 6,882,627, US 7,508,755 and US 7,872,973, US 7,406,260, US 7,423,962
Matter: Investigation and an expert report for '627, '755, and '973. Claim

Constructions' expert report for '260 and '962 re: Claim Construction, MPLS Routing, Packet Switching, Queuing, January 2021.

Case Name: **TCL Communication** Technology Holding
Client: Cozen O'Connor
Patent No: US 6,058,304 and US 7,139,591
Matter: IPR declarations and deposition re: Wireless telephony, handheld wireless device architecture, March 2020.

Case Name: **Polycom**, Inc. v. Direct Packet Research, Inc.
Client: Oblon (PTAB) and Pepper Hamilton (Federal)
Patent No: US 7,710,978, US 7,773,588, and US 8,560,828
Matter: Six PTAB IPR declarations, multiple depositions. Expert reports for federal court, deposition re: Firewall and secured multimedia communications, June 2019.

Case Name: **Avago Technologies** v. Netflix
Client: Hopkins Carley
Patent No: US 8,572,138, US 9,319,289, and US 9,497,264
Matter: An investigation, technology analysis re: Video Streaming, distributed network services, September 2019.

Case Name: Guest Tek v. **Nomadix** - Case No. T-448-17 Federal Court in Canada
Client: Knobbe Martens, and ROBIC LLP
Patent No: CA 600,760, CA 2,750,345, CA 2 790 354, and 2 817 932. US8811184, US9154435, US9531640, US9871738
Matter: An investigation, multiple non-infringement, and invalidity expert reports, **court testimony at trial** Canadian court re: wireless and communications access control, June 2019.

Case Name: **Microsoft** V. Uniloc USA, IPR
Client: Klarquist Sparkman LLP
Patent No: US8,724,622
Matter: An IPR expert report re: An IPR expert report (very similar to Apple, Facebook WhatsApp expert reports), 2019.

Case Name: **Unified Patents** v. FireNet Technologies LLC, IPR
Client: Unified Patents
Patent No: US 8,892,600
Matter: An IPR expert report re: Access and security of data networking attached devices, Network Firewall, Proxy Server, Network security, 2019.

Case Name: **Guaranteed Rate**, Inc. v. Netrix
Client: Novack and Macey LLP
Matter: An expert report, a deposition re: MPLS and broadband architectures using Cisco equipment, 2019.

Case Name: Citrix Systems, Inc. v. **Avi Networks**, Inc.,
Case No. 17-1843-LPS (D. Delaware)
Client: Gibson, Dunn & Crutcher LLP
Patent No: US 9,148,493, US 8,631,120
Matter: Expert declarations for PTAB IPR re: Declaration for Claim Construction brief, depositions, 2019.

Case Name: **Apple** Inc. v. VoIP-Pal.com, Inc.; Inter Partes Review
Client: Erise IP PA.
Patent No: US 9,537,762; US 9,813,330; US 9,826,002; and US 9,948,549
Matter: Expert declarations re: VoIP and network communications, 2019.

Case Name: **Ticket Network**, Inc., and Ticket Software LLC v. CEATS, Inc.
Client: Fish & Richardson P.C
Patent No: US7,548,867, US7,640,178
Matter: IPR Expert reports, deposition re: Web reservation over the Internet and database consistency over multiple reservation instances, 2017-2018.

Case Name: **Amazon** Inc v. Hera Wireless
Client: The WEBB Law Firm
Patent No: US7,454,234, US7,369,878, and US7,873,389
Matter: Research, analysis, and drafting an expert report (not filed) re: Wireless technologies, Wi-Fi 802.11 a/b/n, MIMO, 2018.

Case Name: **ATEN** International Co., LTD v. Uniclass Technology Co., LTD
Client: Mei & Mark LLP (/ Devlin Law Firm LLC)
Patent No: US6,957,287, US7,472,217, US7,640,289 and US8,589,141
Matter: Several infringement expert reports. Deposition and **court testimony at trial** before a jury. Patents related to KVM switch architectures, Bluetooth. Network communications, 2016-2017.

Case Name: **Apple** V. Uniloc USA, IPR2017-01804
Client: Stern Kessler Goldstein Fox
Patent No: US8,724,622
Matter: An IPR expert report re: An IPR expert report (the same as for Facebook WhatsApp), 2017.

Case Name: IPR for **Huawei** and RPX
Client: Wolf Greenfield LLP
Patent No: US6,738,378 & US6,252,848
Matter: An IPR expert report re: Patents in network communications. TCP/IP, Internet Protocols, 2017.

Case Name: **LG** and **Huawei** v. Intellectual Venture

Client: Finnegan, Henderson, Farabow, Garrett &Dunner, LLP
Patent No: US8,994,433
Matter: An IPR expert report re: Wireless Mobile Devices. An IPR expert report, (The same as for Facebook WhatsApp), 2017.

Case Name: **T-Mobile** v. Barkan Wireless Access Technologies LP.
Client: Perkins Coie LLP
Patent No: US8,559,369 and US9,042,306
Matter: An expert for T-Mobile in a single IPR for the two patents re: Wi-Fi Tethering in broadband cellular data, 2017.

Case Name: **Facebook**, Inc., **Instagram** LLC v. Skky, LLC
Client: Cooley, LLP
Patent No: US8,892,465, US9,118,693, US9,124,717, US9,124,718, US9,203,870, US9,203,956, US9,215,310 and US 9,219,810,
Further IPRs for Patent Nos. US8,892,465, US9,118,693, US9,124,717 and US9,124,718
Matter: An expert for defendant Facebook, IPR, and CBM expert reports, Deposition testimony re: Patents related to cellular systems using OFDM, 2017.

Case Name: **Facebook**, **WhatsApp**. V. Uniloc USA., Inc.
Client: Cooley, LLP
Patent No: US8,199,747, US8,995,433, US8,243,723, and US 7,535,890.
IPR2017-01524, IPR2017-01365, IPR2017-01523
Matter: Expert for defendant Facebook. Wrote IPR expert reports. Testified in depositions re: Mobile wireless. Patents related to Unified Communications. VoIP, Cellphone Short Messages, SMS, MMS, Chat app, 2016.

Case Name: **YMax** Corporation v. Focal IP, LLC
Client: Akerman LLP
Patent No: US7,764,777, US8,155,298, US8,724,622 and US 8,457,113
Matter: Expert for the defendant and IPR petitioner, YMax Corp., Wrote three expert reports for the PTAB and testified in depositions regarding Patents related to telecommunications architecture, PSTN, and VoIP.
Streaming media, 2016.

Case Name: **FedEx** (RPX) v. Intellectual Ventures IPR2017-00729, IPR2017-00741
Intellectual Ventures, LLC and Callahan Cellular LLC v. FedEx Corp.
Client: Finnegan, Henderson, Farabow, Garrett & Dunner, LLP
Patent No: US6,633,900 and US8,494,581
Matter: Wrote two IPR expert reports and testified in depositions re: Patents related to mobile devices over wireless networks. Cellular systems, 2016 – 2017.

Case Name: **Patentmarks Communications**, LLC v. Internap Corporation

Client: Warner Norcross & Judd
Patent No: US6,016,307, US6,473,404, US6,144,641, US8,400,926 and US9,036,499
Matter: Defendant's expert. Wrote a claim-construction declaration.
re: Patents related to multi-protocol telecommunications routing optimization. Routing and packet switching, 2016.

Case Name: **Facebook** IPR PTAB v. Windy City
Client: Cooley, LLP
Patent No: US8,458,245, US8,407,356, US8,473,552 and US8,694,657 (IPR2016-01157, IPR2016-01158, IPR2016-01159, IPR2016-01156)
Matter: The expert for petitioner Facebook, Inc. wrote an IPR expert report for the PTAB, and testified in depositions. Re: Patents related to network communications. Mobile wireless (SMS on WhatsApp), 2016.

Case Name: **RPX** IPR PTAB filing.
Client: Winston & Strawn
Patent No: US8,788,090
Matter: Expert for petitioner RPX. Wrote an IPR expert report for the PTAB
re: Patents related to network communications, 2016.

Case Name: **ServiceNow**, Inc. v. BCM Software, Inc.
Client: Cooley, LLP
Patent No: USPTO PTAB IPR Cases: IPR2015-01555; IPR2015-01211; IPR2015-01631; IPR2015-01601. CBM Case CBM2015-00170 ,2:14-CV-0090, US8,646,093; US8,646,093 (CBM as well), US 7,617,073; US8,674,992; US6,816,898
Matter: Expert for the defendant and IPR petitioner, ServiceNow, Inc., wrote IPR and CBM expert reports for PTAB wrote expert reports for claim constructions. Testified in depositions re: Patents related to the network. Management, 2015.

Case Name: **ServiceNow**, Inc. v. Hewlett-Packard Company
Client: Cooley, LLP
Patent No: USPTO PTAB IPR Cases IPR2015-00707; IPR2015-00716; IPR2015-00631; IPR2015-00717; IPR2015-00707. CBM, CBM2015-00108; CBM2015-00077US7,925,981; US7,945,860; US7,392,300; US7,027,411; US7,925,981, US7,945,860 (CBM); US7,925,981 (CBM)
Matter: The expert for the defendant and IPR petitioner, ServiceNow, Inc., wrote IPR and CBM expert reports for the PTAB Cases IPR2015-00631 and IPR2015-00717. Testified in depositions. Re: Patents related to Web services and Internet technologies. Internet protocols, routing, and switching 2015.

Case Name: SNMP Research International, Inc., et al. v. **Avaya**, Inc.
Client: Holland & Knight, LLP

Matter: Expert for defendant Avaya, Inc., wrote expert reports. Provided deposition testimony regarding network management for IP-PBX. Internet Protocols, TCP/IP, Network Management, SNMP, 2015.

Case Name: Qurio Holdings v. **DirectTV** Case No.15-cv-01986-HSG (N.D. CA), and 3 IPR petitions IPR2015-02005, IPR2015-02006, IPR2015-02007
Client: Perkins Coie, LLP
Patent No: US7,787,904, US8,102,863 and US8,879,567
Matter: Expert for the defendant and IPR petitioner, DirectTV, wrote three IPR expert reports re: Mobile device control, WAN to wireless LAN gateway, wireless LAN gateway, 2015.

Case Name: **FedEx** Corporation and FedEx Corporate Services, Inc., Petitioner v. Loramax LLC, Patent Owner
Client: Finnegan, Henderson, Farabow, Garrett & Dunner, LLP
Patent No: US5,689,642 and US5,513,126
Matter: The expert for the defendant and IPR petitioner FedEx Corporation wrote an IPR expert report re: Improving communication between devices or stations. (Plaintiff dismissed the case without IPR filed) 2015.

Case Name: Expert for **GT&T**
Client: McGuire Woods, LLP
Matter: An expert for a cellular telecommunications company re: Analysis of cellular network and phone systems, 2015.

Case Name: MD. Security Solutions, LLC v. **RPX**
Client: Wolf Greenfield, LLP
Patent No: US7,864,983
Matter: Expert for IPR petitioner RPX. Wrote an IPR expert report re: Network communications for alarm systems, 2015.

Case Name: **IXI Mobile** v. Apple Inc. 14-cv-7954 SDNY
Client: Simpson Thacher & Bartlett, LLP
Patent No: US7,295,532, US7,426,398, US7,016,648, and US 7,039,033
Matter: Searched for prior art, prepared invalidity claim charts re: Invalidity claim charts related to iPhone, iPad, and others, Apple's mobile wireless products and technologies, 2015.

Case Name: USPTO Office Action App. 13/432.775 (**NetAirus**)
Client: Hovey Williams, LLP
Patent No: Patent App. 13/432.775
Matter: A short declaration for an office action re: Desktop computer configuring the system, 2015.

Case Name: Spherix v. **Cisco Systems** 1:14-cv-393-SLR (D. Del.)
Client: Kirkland & Ellis, LLP

Patent No: US RE40467, US 6,697,325, US 6,578,086, US 6,222,848, US 6,130,877, US 5,970,125, US 6,807,174, US 7,397,763, US 7,664,123, US 7,385,998, US 8,607,323 US 6,879,594, US 6,661, 788, US 6,233,245, US 6,684,241, US 6,661, 788, US 6,879,594
Matter: Searched for prior art, invalidity claim charts, and investigation re: Network communication, routing, and switching technologies, 2014.

Case Name: Bockstar Technologies, LLC v. **Cisco Systems**, Inc.
Case No. 1: 13-cv-2020-SLR (D. Del.)
Client: Desmarais, LLP
Patent No: US 5,732,080, US 6,069,895, US 6,233,245, US 6,636,508, US 6,684,241, US 6,778,653
Matter: Research and analysis of the prior art re: Network communication, routing, and switching technologies, TCP/IP. Management of bandwidth in a data communication network. Distributed route servers, 2014.

Case Name: **Cisco Systems**, Inc. - USPTO Inter Partes Review (IPR)
(the petition was not filed)
Client: Armstrong Teasdale, LLP
Patent No: US 5,583,862
Matter: Expert for petitioner Cisco Systems, Inc. Wrote invalidation expert reports for the USPTO PTAB re: Patent-related to telecommunications, routing, and switching technologies. Routing Switching, Network Communications, packet switching, Internet Protocols, TCP/IP protocol suite, 2014.

Case Name: **Mobotix** - Inter Partes Review IPR2015-00093
Client: Fish Richardson, LLP
Patent No: US 6,975,220
Matter: Expert for petitioner Mobotix. Wrote invalidation expert reports for the USPTO PTAB regarding the USPTO Inter Partes Review (IPR) patent. related to network architecture, network communications protocols, and architectures, 2014.

Case Name: Brilliant Optical Solutions, LLC v. **Comcast**
District of Colorado, Case no. 1:13-CV-00886-REB-KMT and USPTO Inter Partes Review (IPR)
Client: Faegre Baker Daniels, LLP
Patent No: US 5,555,478
Matter: Expert for the defendant and IPR petitioner, Comcast. Wrote invalidation expert reports for the USPTO PTAB IPR (Plaintiff dismissed the case without filing IPR) regarding patents related to routers' switching fabric architecture. Routing, packet Switching, 2014.

Case Name: Brixham Solutions, LTD, vs. **Juniper Networks**

3:13-cv-00616-JCS (N.D. CA) and USPTO IPR 2014-00425 and IPR 2014-00431
Client: Irell & Manella, LLP, CA
Patent No: US7,940,652 and US7,535,895
Matter: Wrote two invalidation expert reports for the USPTO PTAB. Testified in depositions. re: expert for the defendant and IPR petitioner Juniper Networks, a patent related to routers' switching fabric architecture, and a patent related to Ethernet pseudo-wire VPN. Routing Switching, Network Communications architectures, and protocols, 2014.

Case Name: **Ericsson** v. Samsung (I.T.C.), **ITC Investigation** NO. 337-TA-862
Client: McKool Smith (Dallas, TX)
Patent No: US6,772,215, US6,519,223
Matter: Wrote several expert reports. Testified in depositions. **ITC court testimony at trial** re: Patents related to cellular systems, CDMA, GSM, UMTS, LTE, and wireless network communications. Mobile Systems and Cellular Networks, 2013.

Case Name: **Comcast Cable** Communications, LLC et al. v. BT. Americas, Inc.
3:12-cv-01712-M (ND Tex.)
Client: Keker & Van Nest, LLP
Patent No: US7,142,508, US5,638,516
Matter: Analysis, claim terms, and Markman claim construction re: MPLS technology, Routing Switching, Network Communications, 2013.

Case Name: **NetAirus** Systems LLC v. Kappos
USPTO 1:12-cv-00709-AJT-JFA, 1:12-cv-01207-AJT-TRJ
Client: Williams, LLP (KS)
Patent No: US5,280,583, US5,589,849, US2005/0110780 A1
Matter: Wrote a declaration regarding claim amendment regarding video conference application program, unified communications, streaming video, and VoIP, 2013.

Case Name: e-Watch, Inc. v. **Mobotix** Corp. USPTO Reexamination
Client: Slater&Matsil, LLP (Dallas, TX)
Patent No: US6,970,183
Matter: Expert for the petitioner, Mobotix Corp. Wrote an invalidity declaration - USPTO IPR2013-00255 re: A wireless monitoring system, wireless networking, wire-line networking, 2013.

Case Name: **Teradici's** - USPTO Re-examination
Client: Nixon Peabody (Chicago, IL)
Patent No: US6,012,101, US5,604,509, US5,577,205
Matter: Wrote an invalidity expert report re: Patents related to network communications, remote KVM, 2012.

Case Name: Wassink v. **ACS** (Affiliated Computer Systems, Inc.)

8:11-cv-00554-CJC-MLG (CD, CA)
Client: Morgan, Lewis & Bockius, LLP (Irvine, CA)
Matter: Wrote an expert report re: Cisco certification of IT personnel, Network Communications, 2012.

Case Name: **HumanEyes** Technologies, LTD v. Sony Electronics (ITC.)
1:12-cv-00398 (D. Del.)
Client: Kenyon& Kenyon, LLP
Patent No: 6,665,003, 7,4 77,284
Matter: Investigation of patents and related technologies re: Software related to mobile devices, video streaming including panoramic imaging, mobile wireless devices, 2012.

Case Name: Realtime Data v. **Morgan Stanley** et al.
1:11-cv-6696,-6701,-6704 (S.D. N.Y.)
Client: Skadden, Arps, Slate, Meagher & Flom LLP (New York, NY)
Patent No: 7,417,568, 7,714,747, 7,777,651
Matter: Wrote a rebuttal to an infringement report, was deposed re: Technology related to a network communication protocol, TCP/IP Protocols, streaming media, and QoS 2012.

Case Name: Maxim Integrated Products, Inc. v. **VanguardBank** and PNC Bank
Client: Finnegan, Henderson, Farabow, Garrett & Dunner, LLP
Patent No: US5,940,510, US5,949,880, US6,105,013, US6,237,095
Matter: Investigation of prior art re: Patents litigation related to virtual private networks (VPN) and secure service layer (SSL), Network Security, 2012.

Case Name: **M2M Solutions**, LLC v. Sierra Wireless, CA No. 12-030-RGA (D. Del.)
Client: Foley & Lardner, LLP
Patent No: US7,583.197, US8,094,010
Matter: Investigation of the prior art, claim charts re: Patents related to wireless and mobile devices in cellular systems, 2012.

Case Name: **Site Update** Solutions, LLC v. Accor North America, Inc., et al.
4:11-CV-03306-PJH (ND CA), Ipr2016-01235,
CBS Corp., 15-1448, 13-1458, 5:11cv3306
Client: Goldstein & Lipski, P.LLC (Houston, TX)
Patent No: US RE40.683, US 6,253,198
Matter: A declaration re: Patent litigation related to Internet web technologies and e-commerce, 2012.

Case Name: **EIT** Holdings LLC V. Netflix and Barnes & Noble Inc.
Client: Goldstein & Lipski, P.LLC (Houston, TX)
Patent No: 5:11-CV-02466; 2012 WL 728387, 1:10-CV-01081,
2011 WL 7149251,2011 WL 7149248
Matter: Expert report re: Patent-related to Internet and Web technologies, 2012.

Case Name: **Proxyconn**, Inc. v. Microsoft et al. 8:11-cv-1681-DOC-AN (CD CA)
Client: Russ, August & Kabat (Los Angeles, CA)
Patent No: US6,757,717, US6,370,646
Matter: Investigation of technology was deposed regarding speed data accessing in
communication networks. Bandwidth and network access, 2012.

Case Name: **Walker Digital** v. Google 1:11-cv-00318-LPS (D. Del)
Client: Greenberg Traurig, LLP
Patent No: US5,884,272, US5,884,270, US5,824,082, US 5,136,501, US5,832,497
Matter: Invalidity analysis, investigation of patents and related technologies re: Internet and e-commerce technologies, network communications, 2012.

Case Name: **BackWeb** v. HP and IBM 4:10-cv-04311-PJH
Client: Hosie Rice, LLP (San Francisco, CA)
Matter: Investigated patents and related technologies re: Patent litigation related to network communication, bandwidth allocation, 2012.

Case Name: **High Point** SARL v. Sprint Nextel 2:09-CV-2269 (Kansas City, KN)
Client: Hogan Lovells US, LLP
Patent No: US5,884,272, US5,884,270, US5,824,082, US5,136,501, US5,832,497
Matter: Search for invalidity prior art re: Patents related to mobile wireless and cellular systems, 2011.

Case Name: Emblaze v. **Apple** 1:10-cv-5713 -PKC (SDNY.)
Client: Greenberg Traurig, LLP
Patent No: US6,389,473, US5,841,432, US5,751,968, US6,138,147, US5,610,841, US6,397,230, US5,579,239, US5,928,330, US5,987,510, US6,014,694, US6,151,632, US5,864,682, US5,659,539
Matter: Investigation of patents, invalidity analysis re: Computer network protocol for multimedia applications. Network Communications, Internet Protocol, TCP/IP, SMS, MMS Streaming Video, 2011.

Case Name: **Motorola Mobility**, Inc. v. Microsoft 1:10-cv-24063-MORENO (SD, FL)
Client: Ropes & Gray, LLP (New York, NY)
Patent No: US7,024,214, US7,493,130, US7,725,584, US6,219,694, US5,406,643, US7,275,116
Matter: Wrote an invalidity expert report and wrote a response to an infringement report. Testified in depositions 7/21/2011 re: Patent litigation related to wireless and mobile devices, including smartphones and tablets, 2011.

Case Name: **EIT** v. YELP! Inc. 3:10-cv-05623-WHA (ND CA)
Client: Goldstein & Lipski, P.LLC (Houston, TX)
Patent No: US5,828,837, US6,249,868, US6,594,765, US6,081,786,

US6,188,988, US7,298,271, USRE40,683, US8,060,504
Matter: Wrote an expert report re: Patent litigation related to Internet and Web technologies, network communications, 2011.

Case Name: **Software Research**, Inc. v. HP 3:11-cv-1676-JSW (ND, CA)
Client: Hosie Rice, LLP (San Francisco, CA)
Patent No: US7,231,606, US7,757,175
Matter: Wrote a declaration 7/15/2011 re: Patent litigation related to Internet web technologies and e-commerce, network communications, 2011.

Case Name: Yipes v. **Wyckoff Medical** Center AAA No. 74494 Y 00849 10 NOLG
Testimony at arbitration trial
Client: David N. Hoffman (Brooklyn, NY) and S. Robert Diener (Berkeley, CA)
Matter: Testified in depositions. Testified before an arbitrator regarding network communications infrastructure, Unified Communications, 2011.

Case Name: HP v. **Alacritech** USPTO Interference No. 105,775
Client: Silicon Edge Law Group, LLP
Patent No: US6,246,683, US6,226,680, US6,697,868
Matter: Wrote an expert report, an expert declaration, a response expert report, and a reply expert report. Testified twice in depositions re: USPTO Interference: Patents related to network communications protocols, TCP/IP Protocol Suite, Internet Protocol, 2011.

Case Name: Fortinet v. **Palo Alto Networks** 5:09-cv-00036-RMW (ND, CA)
Client: Durie Tangri, LLP (Palo Alto, CA)
Patent No: US7,519,990, US1,111,111, US7,376,125, US7,177,311, US6,930,978
Matter: Wrote an invalidity expert report. Testified in depositions. re: Patent litigation related to network communications filtering, routing, packet switching, 2010.

Case Name: **Backweb** Technologies v. Microsoft 4:09-cv-01224-CW (ND, CA)
Client: Hausfeld, LLP (San Francisco, CA)
Patent No: US6,374,289, US6,493,758, US6,539,429, US5,913,040, US6,317,789, US6,539,429, US7,512,943, US7,139,815, US7,216,180, US6,920,110, US7,325,068
Matter: Wrote an expert report re: Patent litigation related to Data Communications, network communication bandwidth allocation, routing, packet switching, 2010.

Case Name: **Linksmart** Wireless v. Cisco Systems, T-Mobile
4:09-cv-01224-CW (ND, CA), 2:08-cv-00385,2:09-cv-00026
Client: August & Kabat (Los Angeles, CA)
Patent No: US6,779,118

Matter: Wrote an expert report. Testified in a deposition. re: Patent litigation related to network communication, Internet Protocols, dynamically redirecting and filtering Internet traffic, 2010.

Case Name: Zand v. **Armin** 1-09-FL-149015 - **Court testimony at trial**
Client: Law Offices of Rod Firoozye (Palo Alto, CA)
Matter: Wrote an expert report. Testified in a deposition. Testified in court re: iPhone email system. Email origination in iPhone, desktop computer, and laptop, 2010.

Case Name: **Teleconference System** v. Tandberg
3:10-cv-01325-JSW (ND, CA)
Client: The Simon Law Firm, PC (St. Louis, MO)
Patent No: US6,980,526
Matter: Investigated patent and related technologies regarding network protocols in
the teleconference system. Video Teleconferencing, Video streaming, 2010.

Case Name: F5 Networks v. **A10 Networks** 2:10-cv-00654-MJP (WD WASH)
Client: Finnegan, Henderson, Farabow, Garrett &Dunner, LLP
Patent No: US7,102,996, US7,395,349, US6,311,278
Matter: Invalidity investigation, a search for prior art re: Patents related to network communications, firewalls, and load balancing, 2010.

Case Name: **IP Innovation** v. Google 2:07-cv-503-RRR (ED TX)
Client: Niro Scavone Haller & Niro
Patent No: US5,276,785, US5,675,819
Matter: Wrote an infringement expert report and a rebuttal expert report re: Patent litigation related to Internet and Web technologies, 2009.

Case Name: **Trace Wilco**, Inc. v. Cisco 5:09-cv-000810-SI
Client: Niro Scavone Haller &Niro
Patent No: US6,158,011, US6,061,796, US7,325,063, US5,796,948
Matter: Investigated patents and related technologies regarding Network communications protocols. Routing, switching, VoIP, and video streaming, 2009.